MOHITHA BANDI

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PROFESSIONAL SUMMARY

Data Science and Web Development professional with hands-on experience building and deploying AI-powered applications and scalable web platforms. Proficient in Python, SQL, and JavaScript with expertise in AI/ML development, full-stack engineering, predictive modeling, and data-driven problem solving. Experienced in translating user requirements into technical solutions and integrating AI/ML models into interactive applications, cloud environments, and API-driven systems. Strong background in NLP, LLMs, RAG, Generative AI, and Computer Vision, with a keen interest in AI evaluation, observability, and agentic systems.

PROFESSIONAL EXPERIENCE

Data Engineer Intern | **Opplane**, California *Jun 2025 – Dec 2025*

- Designed and deployed Python-based web scrapers for U.S. and Singapore markets using Scrapy, Selenium, Playwright, and XML sitemaps, improving data acquisition efficiency and accuracy.
- **Automated** ETL-style data collection workflows using API integrations, ensuring timely, reliable, and scalable data pipelines for downstream analytics.
- Collaborated with cross-functional teams using Bitbucket and Docker for version control, containerization, and CI/CD workflow automation.

Data Analytics Intern | **Infotact Solutions**, Bangalore *Mar 2025 – May 2025*

- Built a Power BI dashboard for real-time cryptocurrency price forecasting and risk analysis; performed sentiment analysis using BERT and TextBlob on live BTC data.
- Applied ARIMA, LSTM, and Random Forest models for trend prediction, improving forecasting accuracy and enabling data-driven decision-making.

AI-ML-DS Intern | **BlackBuck (IIDT & APSCHE)**, Bangalore *May 2024 – Jul 2024*

- **Deployed** and optimized ML models in Python, achieving a 25% increase in prediction accuracy and reducing data processing time by **30%.**
- Collaborated with teams on model deployment and performance tuning; recognized as Top Performer for contributions to project outcomes.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **Data Engineering:** ETL Pipelines, Data Cleaning, Data Transformation, Data Modeling, REST APIs, Data Pipeline Automation
- **AI/ML & Data Science:** Scikit-learn, PyTorch, NLP, LLMs, RAG, LlamaIndex, Generative AI, Computer Vision
- **Data Visualization & BI:** Power BI, Excel, Pandas, NumPy, Matplotlib, Seaborn, Dashboard Development
- **Web Scraping:** Selenium, Scrapy, Playwright, Beautiful Soup
- **Frameworks & Tools:** Flask, Django, FastAPI, Streamlit, React.js
- **Databases & Cloud:** MySQL, SQLite, ChromaDB, AWS, Docker
- **Developer Tools:** Git, Bitbucket, VS Code, Jupyter Notebook, Google Colab

EDUCATION

B.Tech in Computer Science (Specialization: Data Science) *2022 – 2026*

Woxsen University, Hyderabad, Telangana, India | Percentage: 86%

PROJECTS

Predicting Mental Health Vulnerabilities from Behavioral & Demographic Patterns *Dec 2025 – Mar 2026*

Tech Stack: *Python, Scikit-learn, XGBoost, Deep Autoencoders, SHAP, Flask, SQLite, HTML/CSS/JavaScript, Explainable AI*

- Designed a privacy-preserving web application predicting mental health vulnerability using a hybrid architecture of Deep Autoencoders for feature learning and XGBoost for classification, achieving **83–85%** accuracy.
- Integrated SHAP (Explainable AI) to identify key risk drivers and protective factors, ensuring model transparency and actionable insights.
- Built a full-stack prototype with Flask backend, SQLite database, and responsive HTML/CSS/JS frontend for anonymous self-assessment; optimized for real-time inference under **2** seconds, applicable to corporate wellness and public health programs.

A Smart System for Personalized Interior Design & AR Try-On *Jul 2025 – Oct 2025*

Tech Stack: *Python, PyTorch, Scikit-learn, ResNet, YOLOv8, Stable Diffusion, OpenCV, WebXR, React.js, FastAPI, SQLite, Generative AI*

- Addressed the challenge of **90%** of users struggling to visualize furniture before purchase by building an AI-powered try-before-you-buy platform with 3D AR visualization.
- Built an ETL pipeline, trained ResNet for room classification, and YOLOv8 for furniture detection.
- Integrated Generative AI for personalized recommendations and layout optimization.
- Delivered a smart **try-before-you-buy** feature with 3D visualization and user feedback integration to enhance decision-making.

Deep Chat: Customizable Chatbot with API & LLM Integration *Jan 2025 – Apr 2025*

Tech Stack: *Python, Transformers, LangChain, Streamlit, LoRA, ChromaDB, Ollama, Tesseract OCR, RAG*

- Designed and deployed an LLM-powered conversational assistant using Retrieval-Augmented Generation (RAG), applying LoRA-based fine-tuning and model optimization techniques to reduce model size by 70% while maintaining performance quality.
- Enhanced chatbot performance by optimizing semantic search and retrieval pipelines with SentenceTransformers and ChromaDB, reducing response latency by 30% and improving user interaction experience.
- Developed a scalable LangChain-based application supporting interactions across text, PDFs, scanned documents, and images through OCR integration, expanding accessibility and enabling seamless multi-format knowledge retrieval.

AI-Enabled Phishing Detection *Jan 2025-Feb 2025*

Tech Stack: *Python NLP, Machine Learning, Flask, Scikit-learn*

- Developed an AI-powered phishing detection system to identify malicious emails and fraudulent patterns using machine learning techniques.
- Applied text preprocessing, feature extraction, and classification models to improve phishing detection accuracy.
- Built an interactive interface for real-time email threat analysis and prediction.

Credit Card Fraud Detection *Nov 2024*

Tech Stack: *Python, Scikit-learn, Pandas, Flask*

- Built a machine learning model to detect fraudulent credit card transactions using anomaly detection and classification techniques.
- Performed data preprocessing, feature engineering, and model evaluation on imbalanced datasets.
- Improved fraud detection performance using predictive analytics and data-driven validation methods.

Natural Disaster Analysis & Prediction System *Aug 2024 – Dec 2024*

Tech Stack: *Python, Machine Learning, Flask, REST API Integration, Pandas, Matplotlib*

- Developed flood forecasting models for India by analyzing historical rainfall and disaster datasets; deployed via Flask with real-time weather API integration.
- Implemented a proactive alert system using monthly and annual rainfall trend analysis, enabling timely risk assessment and disaster management decisions.

An Incremental Bias Selection in Recommendation Systems *Feb 2024 – Apr 2024*

Tech Stack: *Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Web Scraping*

- Developed an algorithm to improve accuracy and fairness in recommendation systems using the Proximity-Impact-Popularity (**PIP**) measure.
- Delivered user-oriented, relevant recommendations independent of user history, reducing algorithmic bias in collaborative filtering models.

CERTIFICATIONS

- Human Computer Interaction -NPTEL
- Big Data Analytics with SQL – Cloudera
- Exploratory Data Analysis for Machine Learning – IBM
- Reinforcement Learning – NPTEL
- Programming Foundations with JavaScript, HTML and CSS – Coursera

ACHIEVEMENTS

- **1st Prize** – Essay Competition "ISRO Future Space Mission – Chandrayaan-4" at NRSC, ISRO, Hyderabad.
- **Top Performer** – Recognized at AI-ML-DS Internship (BlackBuck, IIDT, APSCHE) and Data Analytics Internship (Infotact Solutions).
- HackSavvy-24 Hackathon, MGIT – **Finalist.**

RESEARCH PAPER

AI-Based Fake OTP Detection: A Systematic Review and Proposed Detection Framework

Co-author with Asst.Prof. Vaishali Thakur | International Conference on Sustainable Science and Technology for Tomorrow(SciTech-25)

- Conducted a systematic review of AI-based techniques for detecting fake OTP (One-Time Password) attacks, analyzing existing detection methodologies and identifying research gaps.
- Proposed a novel detection framework leveraging machine learning algorithms to identify and flag fraudulent OTP
- patterns in real time.
- Presented research findings at SciTech-25, an international conference focused on sustainable science and emerging technologies.

LANGUAGES

English (Fluent) | **Hindi** (Intermediate) | **Telugu** (Native)